

MARCOS A. CRUZ BAZAN

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SUMMARY

Computer Science student with proven experience building low-latency socket networking protocols, explainable machine learning risk engines, and cross-platform full-stack architectures. Certified Swift Developer proficient in C++, Python, and relational database systems, with a track record of translating complex low-level mechanics into reliable, high-throughput software

EDUCATION

FLORIDA INTERNATIONAL UNIVERSITY

Expected Graduation: December 2026

Bachelor of Science in Computer Science

MIAMI DADE COLLEGE

Graduated: December 2023

Associate of Arts in Computer Science

Honors & Awards: Presidential Scholars Scholarship Recipient, Member of Phi Theta Kappa Society

PROFESSIONAL EXPERIENCE

FLORIDA INTERNATIONAL UNIVERSITY

Miami, FL

Student Office Associate, Scanning Operations – COB Graduate Enrollment

October 2024 – Present

- **Database Data Integrity:** Streamline and execute document processing requests submitted by program recruiters directly within Salesforce CRM, enhancing data accuracy across student files.
- **Data Life-cycle Management:** Track and execute extensive application life-cycle updates, including test score record matching, program metrics, term transitions, and operational reporting.
- **Cross-Functional Communication:** Serve as a core communication bridge between technical operations recruiters and academic administrators while maintaining absolute compliance with privacy regulations.

TECHNICAL DATA PROJECTS

TruthScore — JPMorgan Chase & Co.

- **Data Modeling & Relational Design:** Engineered full-stack relational database systems and automated data modeling pipelines to ingest, store, and process massive synthetic and public identity data sets.
- **Predictive Analytics & Modeling:** Integrated a transparent, rule-based analytics model that scales dynamically into a lightweight Machine Learning system to evaluate real-time signals and baseline behaviors.
- **Stakeholder Visualization & Communication:** Designed an innovative Explainable AI (XAI) UI dashboard that translates complex, black-box numbers into plain-English metrics, directly allowing non-technical business stakeholders to make data-driven security decisions.
- **Platform Health Monitoring:** Established end-to-end backend API services and structured tracking parameters to continuously monitor the health, latency, and reporting efficiency of the platform.

Socket FTP Protocol Client

- **Low-Level Socket Architecture:** Implemented a bare-metal RFC 959 FTP client directly over raw BSD TCP sockets without third-party networking abstractions or wrappers.
- **Stateful Handshake Control:** Handled stateful multi-step TCP control handshakes, automated binary stream payload transmissions, and parsed raw packet byte flows.
- **Dynamic Port Arithmetic:** Implemented dynamic passive transport (PASV) port decoding via 8-bit left shifts and arithmetic recombining ($P1 * 256 + P2$) to dynamically mount parallel data transmission channels.

FleetFlow Logistics

- **Kinematic Simulation Modeling:** Developed an object-oriented simulation framework in C++ modeling autonomous delivery drone vectors, kinematic acceleration, and freight routing topologies.
- **Spatial Constraint Resolution:** Executed continuous collision-detection solvers and constraint-resolution loops across spatial coordinate systems to evaluate supply chain throughput under volatile compute loads.
- **Algorithmic Runtime Optimization:** Optimized routing lookup tables and object dispatch matrices, ensuring deterministic computation times during multi-agent simulations.

SchemaFlow Studio

- **Visual Canvas Architecture:** Built an interactive node-graph modeling tool in Flutter Web to map entity dependencies, relational cardinalities, and foreign-key constraints directly on an interactive canvas.
- **Automated DDL Compilation:** Engineered an on-the-fly DDL compilation pipeline that transforms visual schema topology into sanitized, production-ready ANSI SQL DDL scripts.

Quantitative Trading Algorithm Engine

- **High-Frequency Data Manipulation:** Utilized specialized cache-friendly Ring Buffers, custom Hash Maps, and self-balancing Red-Black Trees to stream, parse, and represent dense real-time market order-book records under low-latency constraints.
- **Statistical Performance Optimization:** Designed an optimized computational runtime engine in C++, successfully scaling execution lookup search structures from $O(N)$ linear operations down to $O(1)$ constant complexity speeds via direct pointer alignment maps.
- **Macroeconomic Trend Discovery:** Isolated cyclical volatility anomalies correlated with annual federal voting seasons; mathematical modeling confirmed that 80% of identified historical election cycles generated a 35% increase in market volatility, whereas 20% demonstrated clear legislative trading volume compression.
- **Team-Driven Competition:** Collaborated closely with a quantitative team in an intense, fast-paced collegiate programming environment, out-performing peers to secure 3rd Place overall in both the planning and algorithm execution categories.

SKILLS

- **Programming Languages:** Python (Statistical Analysis, AsyncIO, Raw Sockets), C++, Java, Swift, Dart, C
- **Developer Tools:** Azure SQL Database, Canvas API, Git, Salesforce CRM
- **Visualization & Frameworks:** Flutter, SwiftUI, FastAPI, Node.js, Hono API, Scikit-Learn, SQLAlchemy ORM
- **Systems & Networking:** Raw Socket Programming, Zero-Trust IAM, Linux, Docker Containerization
- **Certifications:** App Development with Swift (Issued: May 2024)